Assignment 1: Linear Algebra Foundations and Matrix Decompositions

Preparation for Sessional Examination 1

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Assignment 1

Courses: DS5003 Statistical and Mathematical Methods for Data Science / CS5144 Mathematics for Computational Intelligence
Submission deadline: Tuesday, September 29, 2026 — hand in during class
Raw marks: 45 (to be scaled according to the course assessment scheme)

Purpose

This assignment consolidates the material covered before Sessional Examination 1. You are strongly encouraged to attempt all ten questions before the examination, identify topics that need revision, and finalize your submission afterward. The assignment is not a list of examination questions, but it tests the same underlying concepts and skills.

The assessed scope includes vectors and norms, matrices and rank, dot products and projections, determinants and eigenvalues, diagonalization, singular value decomposition, rank-k approximation, principal component analysis, and a short reflective question connecting the course's approach to a classic argument about the nature of data analysis.

Submission instructions

1. Show the essential mathematical steps. Unsupported final answers alone will not receive full credit.
2. Write your solutions by hand, clearly organized and in question order, with your name and section on the first page.
3. Hand in your submission in person during class — this is a physical, paper submission, not a digital one.
4. You may check your arithmetic using a calculator, but the mathematical reasoning and steps must be written out in full, in your own hand.
5. Cite any external source used. If an AI tool assists you in understanding a concept, disclose how it was used; the reasoning and final submission must remain your own work.
6. State any assumption you make.

Question 1 — Norms and the meaning of distance [4 marks]

Let

$$x = \begin{bmatrix} 3 \\ -4 \\ 1 \end{bmatrix}.$$

1. Compute $\|x\|_1$, $\|x\|_2$, and $\|x\|_\infty$.
2. Relative to the query point $q = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, compare $p = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$ and $r = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$:
   - Which point is closer under L_1 distance?
   - Which point is closer under L_2 distance?
3. In one or two sentences, explain why changing the norm can change the meaning of "nearest."

Question 2 — Matrix as a function, column picture, and rank [5 marks]

Consider

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \qquad z = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}.$$

1. Compute Az using the ordinary row-by-column calculation.
2. Recompute it as a linear combination of the columns of A.
3. Determine $\text{rank}(A)$ and identify one dependence relation among its columns.
4. Is A rank-deficient in the usual $\text{rank}(A) < \min(m, n)$ sense? Does it have full column rank?
5. Without calculating a determinant, explain whether $A^T A$ is invertible.

Question 3 — Dot product and cosine similarity [4 marks]

Let

$$u = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \qquad w = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

1. Compute $u \cdot w$.
2. Compute their cosine similarity.
3. What does the sign of the dot product tell you about the angle between them?
4. Can you conclude that the vectors point in exactly opposite directions? Explain.
5. If u is replaced by $10u$, what changes in the raw dot product and what remains unchanged in cosine similarity?

Question 4 — Projection and residual [5 marks]

Let $a = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$.

1. Derive and calculate the scalar coefficient used to project b onto a.
2. Find $\text{proj}_a(b)$.

3. Find the residual $r = b - \text{proj}_a(b)$.
4. Verify numerically that the residual is orthogonal to a.
5. Explain geometrically why this projection is the closest point to b on the line spanned by a.

Question 5 — Determinant, trace, eigenvalues, and eigenvectors [6 marks]

For

$$B = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix},$$

1. Calculate $\det(B)$ and $\text{tr}(B)$.
2. Form the characteristic equation and find both eigenvalues.
3. Find one eigenvector corresponding to each eigenvalue.
4. Verify one eigenpair by direct multiplication.
5. Use the determinant, eigenvalues, or rank to explain why B is invertible.
6. Confirm that the sum and product of the eigenvalues agree with the trace and determinant.

Question 6 — Diagonalization and its limitations [4 marks]

Using the eigenpairs from Question 5:

1. Construct matrices P and D such that $B = PDP^{-1}$.
2. Explain why diagonalization makes powers such as B^{10} easier to understand or compute. You do not need to calculate B^{10}.
3. Is it valid here to replace P^{-1} by P^T? Explain the condition under which that replacement is valid.
4. Give one reason singular value decomposition is more generally applicable than eigendecomposition.

Question 7 — Reading an SVD correctly [5 marks]

A document-term matrix C contains 6 documents and 10 terms, so $C \in \mathbb{R}^{6\times 10}$. We form a rank-2 approximation

$$C_2 = U_2 \Sigma_2 V_2^T.$$

1. Give the dimensions of U_2, Σ_2, and V_2^T.
2. Which singular vectors represent directions in document space, and which represent directions in term space?
3. Does SVD exist even though C is not square?
4. If a software package returns a singular vector with every sign reversed, does this necessarily indicate an error? Explain.
5. Why might a rank-2 representation reveal two latent topics even though no topic labels were supplied?

Question 8 — Truncated SVD, energy, and approximation error [5 marks]

Suppose a matrix has singular values

$$\sigma_1 = 4, \qquad \sigma_2 = 3, \qquad \sigma_3 = 1.$$

1. Calculate the total energy $\sum_i \sigma_i^2$.
2. Calculate the proportion of energy retained by the rank-1 and rank-2 approximations.
3. Calculate the Frobenius-norm reconstruction errors for the rank-1 and rank-2 approximations.
4. If the requirement is to retain at least 95% of the energy, which value of k should be chosen?
5. State precisely what the Eckart–Young theorem guarantees about a truncated SVD.

Question 9 — Principal Component Analysis via SVD [5 marks]

Consider four data points in two dimensions, arranged as rows of a data matrix X:

$$X = \begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 2 \\ 4 & 4 \end{bmatrix}.$$

1. Compute the mean of the rows and construct the mean-centered data matrix X_c.
2. PCA is computed in practice via the SVD of the centered data: $X_c = U\Sigma V^T$. Rather than finding the full SVD by hand, use the shortcut $X_c^T X_c = V\Sigma^2 V^T$: compute $X_c^T X_c$ and find its eigenvalues and eigenvectors.
3. The eigenvector of $X_c^T X_c$ with the larger eigenvalue gives the first principal component direction. State it, and briefly say whether it matches what you would expect from the layout of the four original points.
4. Calculate the proportion of total variance explained by each principal component (the eigenvalue of $X_c^T X_c$ for that component, divided by the sum of both eigenvalues).
5. Some textbooks instead present PCA via eigendecomposition of the covariance matrix $\Sigma_{\text{cov}} = \frac{1}{n} X_c^T X_c$. Without recomputing from scratch, explain how the eigenvectors and eigenvalues of Σ_{cov} relate to those of $X_c^T X_c$ found in part 2. Are the principal component directions the same? What happens to the eigenvalues?
6. Connect this to the "energy" idea from Question 8 — what plays the same role here that the squared singular values played there?

Question 10 — Tukey and the Purpose of Formal Tools [2 marks]

On Day 1, we argued that scaling data-driven systems reliably requires formal mathematical tools rather than ad hoc solutions. Statistician John Tukey, in his 1962 paper "The Future of Data Analysis," argued that data analysis is a broader empirical discipline than mathematical statistics alone — that practical judgment matters as much as formal technique.

Recommended reading: pages 1–7 of the paper (through the end of Section 4, "Sciences, mathematics, and the arts") — link on Classroom. The remaining 60 pages cover specific 1960s statistical techniques not needed here.

In 2–3 sentences, how do these two views relate in the context of this course?

Final self-check before submission

- Have I distinguished rank from matrix shape?

- Have I distinguished a negative dot product from exactly opposite vectors?
- Have I checked that a projection residual is orthogonal to the target direction or subspace?
- Have I kept P^{-1} distinct from P^T unless orthogonality is established?
- Have I identified the roles and dimensions of U, Σ, and V^T?
- Have I defined SVD energy as the sum of squared singular values?
- Have I mean-centered before computing a covariance matrix for PCA?
- Have I interpreted the calculations rather than only reporting them?